

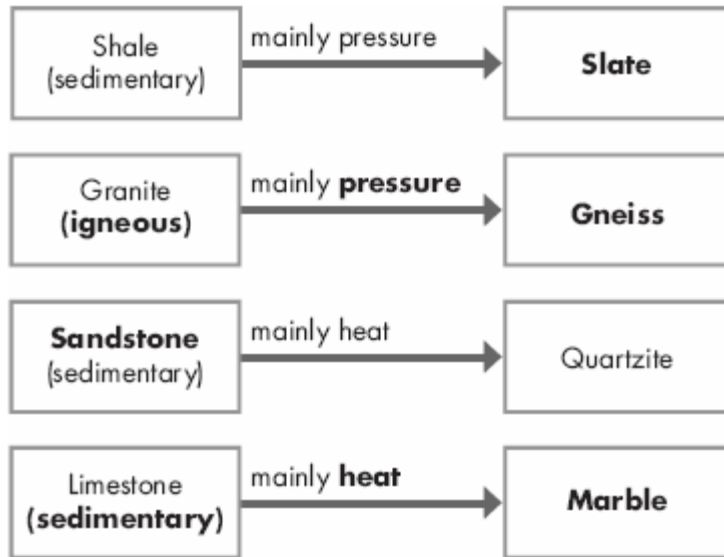
Looking back

1. The crust and the upper mantle.
2. Igneous rocks are formed from molten lava released by erupting volcanoes.
3. Extrusive rocks are igneous rocks that cooled quickly from a molten state above the ground. Intrusive rocks are igneous rocks that cooled from a molten state below the ground; they cool more slowly than extrusive rocks.
4. Answers will vary but may include basalt, obsidian, pumice and scoria.
5. Large crystals are formed when lava cools slowly during intrusive rock formation. Smaller crystals result from faster cooling during extrusive rock formation.
6. Sediments are deposits of weathered rock.
7. Sedimentary rocks can be formed by:
 - cementing of grains of weathered rock
 - deposits of the remains of living things
 - the evaporation of water from a layer of sediment.
8. Test the rock by adding acid; if it is limestone, it will fizz.
9. Some layers are bent or tilted because of the movement of the Earth's layers beneath the surface.
10. Answers will vary. The ripple marks could have been made by water waves on a beach or under the water. If the rippled sediment was buried before being disturbed by rough water or wind, the ripples could be preserved.
11. Answers will vary but should include immense heat and pressure. Igneous and sedimentary rocks are buried beneath the huge weight of soil and rock above; with each kilometre below the surface, the temperature increases by 25 °C. This heat and pressure change the composition of the rock.
12. Parent rock is the rock that another rock has been formed from. For example, limestone is the parent rock of marble.

13. Answers will vary. An example is given.

Type of rock	How it is formed	Special features	Example	Uses
Igneous	Remelting	Crystals	Granite	Headstones
Sedimentary	Weathering and erosion	Layers	Limestone	Buildings
Metamorphic	Heat and pressure	Coloured bands	Slate	Pool table

14.



15. (a) B, E

(b) A, F

(c) C, D

16. Large crystals form when magma cools slowly, while small crystals form when it cools quickly. Granite is an intrusive rock and basalt is an extrusive rock, so granite cooled more slowly than basalt so its crystals are larger.

17. (a) Lustre describes the way light is reflected from a mineral.

(b) Streak is a powdery mark left by a mineral when scraped across a hard surface.

(c) Hardness is determined by trying to scratch one mineral with another. The harder mineral leaves a scratch on the softer mineral.

18. Hardness

19. Scratch one with the other; the one that is scratched is the softer of the two.

20. Flint could be shaped to fit the desired purpose by knocking it with another rock.

21. (a) Hit the flint with another rock, causing it to flake and pieces to shear off.

- (b) It must be harder and not flake or shear when it is hit against the flint.
- 22. Oxygen is most common element in the Earth's crust and is usually found combined with silicon and metal elements.
- 23. to 26. Answers will vary.
- 27. Trace fossils, such as footprints, root and leaf imprints, and feather impressions
- 28. Younger layers may be found beneath older layers if the rock has been lifted and tilted.
- 29. (a) Trace fossil
 - (b) It is evidence of life in the past.
 - (c) No. This one was probably buried in sediment before it could be blown or washed away.
 - (d) The size and weight of the dinosaur and how the dinosaur walked
 - (e) Fossilised bones, teeth, skin impressions, droppings and unhatched eggs